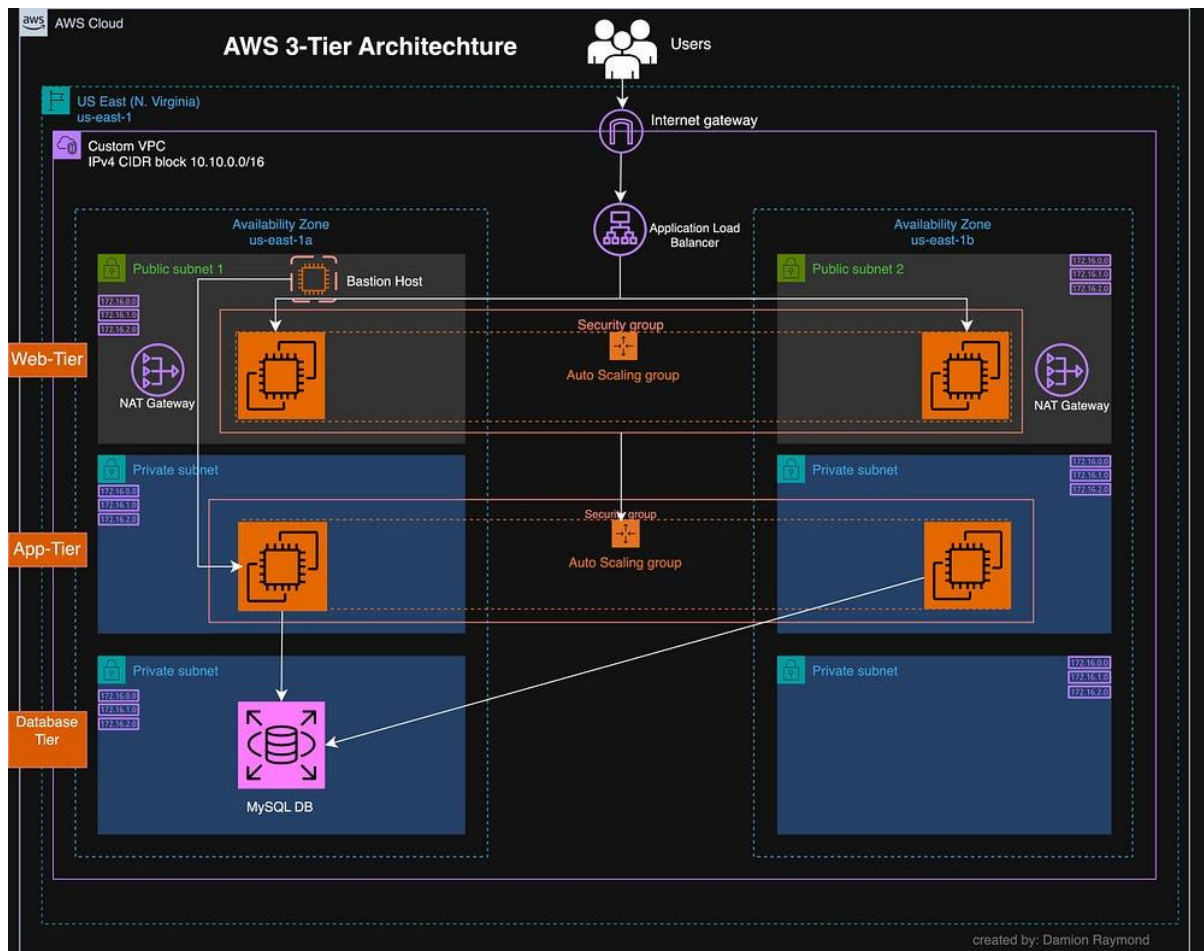


AWS 3-Tier Architecture



A 3-tier architecture is a commonly used software application architecture divided into three distinct tiers (*logical and physical*):

- **Web Tier (Presentation Tier):** The web tier sits at the top of the application. This is where you as a user, interact with the application. It is responsible for collecting information from the user and displaying the content in a graphical user interface (GUI), desktop application, or a website in your web browser.

For example, a webpage written in HTML, CSS, and JavaScript would run at the web tier. The web tier communicates with the application and database tier.

- **Application Tier:** The application tier, aka the logic tier, is where the data processing occurs. Think of it as a bridge between the web tier and the database tier. The application tier is typically developed with various programming languages like Python, Perl, PHP, and Java.
- **Database Tier:** The database tier is where all of the information processed by the application tier is stored and managed. Some common database programs are MySQL, Oracle, and MongoDB.

Benefits of a 3-tier architecture

- **Improved Scalability** — Sudden spikes in traffic are easily handled in an automated fashion with EC2 instances running in an Auto Scaling Group.
- **Improved Reliability** — Deploying servers in two different availability zones ensures that the application will continue to run if one zone goes down due to unforeseen circumstances such as natural disasters or power outages.
- **Increased Security** — By using security groups, we can effectively filter traffic from the web tier to the database tier preventing malicious exploits.

Real time Scenario

A relatively small e-commerce company wants to take advantage of the cloud's scalability, reliability, and improved security by deploying its website on AWS utilizing a 3-tier architecture. The web tier will collect data from users and display content. The application tier will process all incoming data such as items purchased, payment information, and shipping options. The database tier will store all of the data such as transaction data, customer information, and product information.

Solution

- The web tier will include EC2 instances running in an auto scaling group to handle increased traffic during peak sale seasons. These instances will be placed in two public subnets, in two different availability zones allowing public access to our servers.
- The application servers will run on EC2 instances running in an auto scaling group in two private subnets, in two different availability zones. To protect the application tier from being accessed directly from the internet, we will only allow inbound traffic from a web server security group.
- The database tier will utilize a free-tier MySQL RDS database that can easily scale as the platform grows. The database instances will be placed in one private subnet even though we will create two different subnets. Only incoming traffic will be allowed via the application server security group.

By utilizing this 3-tier architecture, this e-commerce company can benefit from reduced overhead, less manual intervention, increased reliability, scalability, and improved security.

Prerequisites

1. AWS Account
2. CLI access

Networking

Creating a VPC

Before we create our first tier, we must first create a network in the AWS cloud that our tiers will have access to. This logically isolated network in the cloud is called a Virtual Private Cloud or VPC. VPCs live in one region. I'll be using the **us-east-1** region. This will be our private network in AWS. You can read more about VPCs in AWS's official documentation

Implementation Steps:

1. Navigate to the VPC console
2. Click "**Create VPC**"
3. Select "**VPC only**" under VPC settings
4. Give your VPC a name in the Name tag section
5. Select IPv4 CIDR manual input

6. We will input our CIDR Block value of **10.10.0.0/16** in the IPv4 CIDR field. This will give us 65,536 IP addresses.
7. We won't be working with IPv6 so leave the No IPv6 CIDR block option selected.
8. The tenancy will be Default
9. Create VPC

Create VPC [Info](#)

A VPC is an isolated portion of the AWS Cloud populated by AWS objects, such as Amazon EC2 instances.

VPC settings

Resources to create [Info](#)

Create only the VPC resource or the VPC and other networking resources.

☒ VPC only

☐ VPC and more

Name tag - *optional*

Creates a tag with a key of 'Name' and a value that you specify.

3tierVPC

IPv4 CIDR block [Info](#)

☒ IPv4 CIDR manual input

☐ IPAM-allocated IPv4 CIDR block

IPv4 CIDR

10.10.0.0/16

CIDR block size must be between /16 and /28.

IPv6 CIDR block [Info](#)

☒ No IPv6 CIDR block

☐ IPAM-allocated IPv6 CIDR block

☐ Amazon-provided IPv6 CIDR block

☐ IPv6 CIDR owned by me

Tenancy [Info](#)

Default

Tags

A tag is a label that you assign to an AWS resource. Each tag consists of a key and an optional value. You can use tags to search and filter your resources or track your AWS costs.

Key

Q Name

X

Value - *optional*

Q 3tierVPC

X

Remove tag

Add tag

You can add 49 more tags

Cancel

Create VPC

Subnets

Subnets are ranges of IP addresses in your VPC. They reside in a single availability zone. You can think of them as blocks in your neighborhood. These subnets will host an autoscaling group of t2.micro instances with Apache installed on each instance. We will create a total of six subnets; two public subnets for our web tier, two private subnets for our application tier, and two private subnets for our database tier.

Public Subnet Creation:

We will create two public subnets with CIDR blocks of **10.10.1.0/24** and **10.10.2.0/24** following the steps below. These subnets will host an autoscaling group of t2.micro instances with Apache installed on each instance.

Steps

1. Navigate to the VPC Dashboard and select “**Subnets**”.
2. If you select the dropdown arrow under Filter by VPC, select the VPC you just created.
3. Select “**Create subnet**”.
4. Choose the VPC you just created under VPC ID.
5. Name your first subnet.
6. Choose an Availability Zone. I’ll use **us-east-1a**.
7. The IPv4 VPC CIDR block value will be **10.10.0.0/16**.

8. We will use **10.10.1.0/24** for the IPv4 subnet CIDR block value.
9. Select “**Add new subnet**”.
10. Follow the above steps to create the next subnet with an IPv4 subnet CIDR block value of **10.10.2.0/24**. Create the next subnet in a different availability zone (AZ) for increased reliability should one site go down, for example, **us-east-1b**. Select “**Create subnet**”.

Private Subnet Creation:

To shield our application and database tier from direct access from the internet, we will create two private subnets for the application tier and two private subnets for the database tier.

Steps

1. Navigate to the VPC Dashboard and select “**Subnets**”.
2. Select “**Create subnet**”.
3. If you select the dropdown arrow under Filter by VPC, select the VPC you just created. Choose the VPC you just created under VPC ID.
4. Name your subnet.
5. Choose an Availability Zone. I'll use **us-east-1a**.
6. The IPv4 VPC CIDR block value will be **10.10.0.0/16**.

7. We will use **10.10.3.0/24** for the IPv4 subnet CIDR block value.

8. Select “**Add new subnet**”.

9. Follow the above steps to create the next subnet for the application tier with an IPv4 subnet CIDR block value of **10.10.4.0/24**. Create the next subnet in a different availability zone (AZ) for increased reliability should one site go down, for example, **us-east-1b**. Repeat the same steps to create the remaining two private subnets for the database tier. CIDR info is listed below. Select “**Create subnet**”.

Note: The CIDR values for the private subnets are listed below.

Application Tier IPv4 subnet CIDR blocks values:

- 10.10.3.0/24
- 10.10.4.0/24

Database Tier IPv4 subnet CIDR block values:

- 10.10.5.0/24
- 10.10.6.0/24

Click on Subnets in the left pane of the VPC Dashboard to verify that all six subnets were created.

Subnets (12) Info

Actions ▾

Create subnet

Q Find resources by attribute or tag

< 1 >

	Name ▾	Subnet ID ▾	State ▾	VPC ▾	IPv4 CIDR
	public-subnet-1	subnet-0bfe558838c8ca011	✔ Available	vpc-08b90d787f5da9326 3tie...	10.10.1.0/24
	public-subnet-2	subnet-0b00d7bdb13567108	✔ Available	vpc-08b90d787f5da9326 3tie...	10.10.2.0/24
	App-private-subnet-1	subnet-08a81f9128214c43b	✔ Available	vpc-08b90d787f5da9326 3tie...	10.10.3.0/24
	App-private-subnet-2	subnet-0a7a3b32c52721062	✔ Available	vpc-08b90d787f5da9326 3tie...	10.10.4.0/24
	database-private-subnet-1	subnet-0e2945cb267d36d40	✔ Available	vpc-08b90d787f5da9326 3tie...	10.10.5.0/24
	database-private-subnet-2	subnet-02b6f2d009ad560e6	✔ Available	vpc-08b90d787f5da9326 3tie...	10.10.6.0/24

Subnet configuration

To ensure that our EC2 instances can be publicly accessed from the internet, we need to configure our two public subnets so that IPv4 addresses can be automatically assigned. We will configure each one separately by following the steps below.

Steps

1. Select the first public subnet by checking the box next to it and then go to **Actions > Edit subnet settings**.
2. Check the box next to Enable auto-assign public IPv4 address and select **Save**.

Edit subnet settings [Info](#)

Subnet

Subnet ID	Name
subnet-0bfe558838c8ca011	public-subnet-1

Auto-assign IP settings [Info](#)

Enable AWS to automatically assign a public IPv4 or IPv6 address to a new primary network interface for an instance in this subnet.

☒ **Enable auto-assign public IPv4 address** [Info](#)

☐ **Enable auto-assign customer-owned IPv4 address** [Info](#)
Option disabled because no customer owned pools found.

Resource-based name (RBN) settings [Info](#)

Specify the hostname type for EC2 instances in this subnet and optional RBN DNS query settings.

☐ **Enable resource name DNS A record on launch** [Info](#)

Repeat the same steps to configure the second public subnet.

Internet Gateways

How can our EC2 instances access the internet from our public subnets? The answer to that question is through an Internet Gateway. Let's create our Internet Gateway.

Steps

1. Select **Internet gateways** in the left pane of the VPC dashboard.
2. A default Internet gateway is created for us but we will create a new one. Select **Create internet gateway**.

3. Name the Internet gateway.
4. Select **Create internet gateway**.

Create internet gateway [Info](#)

An internet gateway is a virtual router that connects a VPC to the internet. To create a new internet gateway specify the name for the gateway below.

Internet gateway settings

Name tag
Creates a tag with a key of 'Name' and a value that you specify.

3tier-IGW

Tags - optional
A tag is a label that you assign to an AWS resource. Each tag consists of a key and an optional value. You can use tags to search and filter your resources or track your AWS costs.

Key	Value - optional	
Name	3tier-IGW	Remove

[Add new tag](#)

You can add 49 more tags.

[Cancel](#) [Create internet gateway](#)

The internet gateway must be attached to a VPC to give the EC2 instances access to the internet so we must attach the internet to the custom VPC we created earlier.

Steps

1. Select the newly created internet gateway and select **Actions > Attach to VPC**.
2. Select the custom VPC we created earlier and select **Attach internet gateway**.

NAT Gateway

For our EC2 instances in our private subnets to have access to the internet securely, we must create a NAT Gateway. This will allow our EC2 instances the ability to access external services outside of our VPC and prevent those external services from accessing our EC2 instances. NAT Gateways are created in specific Availability Zones. Since we have two Availability Zones, we will create a NAT Gateway in each so that if one Availability Zone should fail, we have a second NAT Gateway for our resources to use from a reliability standpoint.

Steps

1. Select **NAT gateways** in the left pane of the VPC dashboard.
2. Select **Create NAT gateway**.
3. Name the NAT gateway.
4. Select the public subnet in AZ1 we created earlier.
5. The connectivity type is **Public**.
6. Select **Allocate Elastic IP**.
7. Select **Create NAT gateway**.

Repeat the steps above to create the second NAT Gateway in the second Availability Zone.

Create NAT gateway [Info](#)

A highly available, managed Network Address Translation (NAT) service that instances in private subnets can use to connect to services in other VPCs, on-premises networks, or the internet.

NAT gateway settings

Name - *optional*

Create a tag with a key of 'Name' and a value that you specify.

The name can be up to 256 characters long.

Subnet

Select a subnet in which to create the NAT gateway.

Connectivity type

Select a connectivity type for the NAT gateway.

☒ Public☐ Private

Elastic IP allocation ID [Info](#)

Assign an Elastic IP address to the NAT gateway.

[Allocate Elastic IP](#)

▶ Additional settings [Info](#)

Tags

A tag is a label that you assign to an AWS resource. Each tag consists of a key and an optional value. You can use tags to search and filter your resources or track your AWS costs.

Key

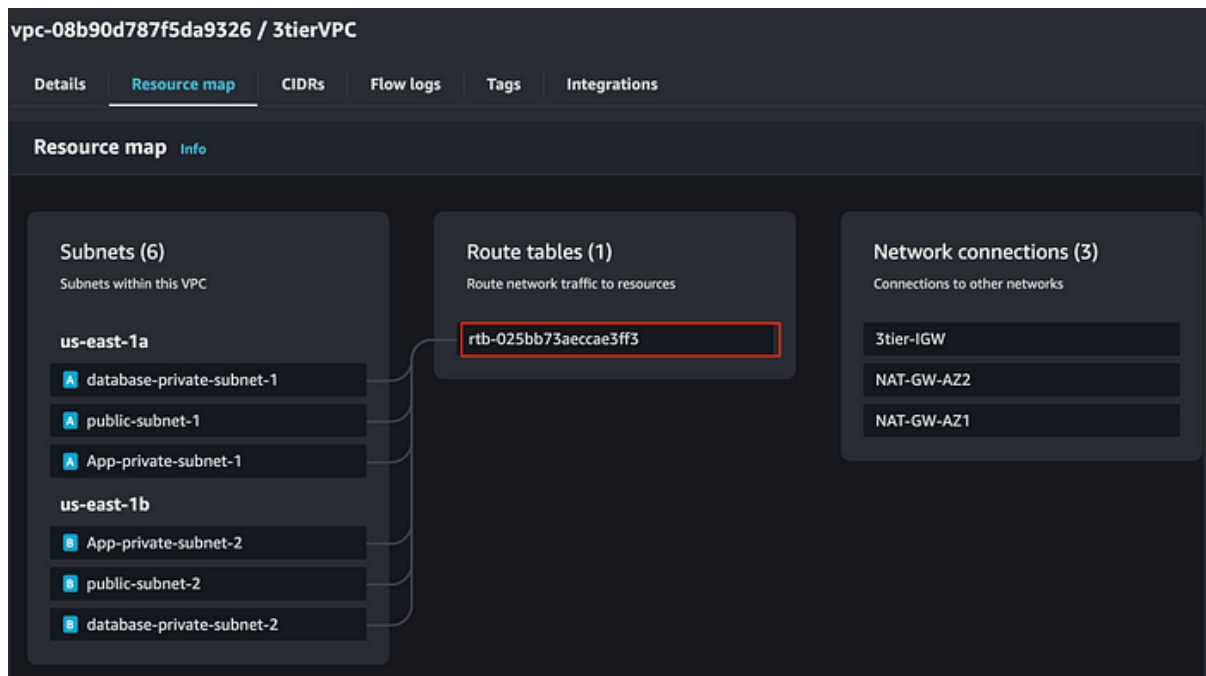
Value - *optional*

You can add 49 more tags.

Routing Configuration

Whenever we created our custom VPC, a main route table is created by default. The main route table is also associated with all of our subnets implicitly. This means that the subnets haven't been

explicitly associated with a route table. We can see evidence of this under our **Resource map**.



Route Tables

There was already a route table created when we created our custom VPC. This is our main route table. We need to edit this route table to enable our public subnets access to the internet through our internet gateway.

Steps

1. Select **Route tables** in the left pane of the VPC dashboard.
2. Select the main route table and then **Actions > Edit routes**
3. Select Add route. Choose **0.0.0.0/0** in the Destination dropdown list. This represents all IPv4 addresses.

4. Select **Internet Gateway** in the Target dropdown list and then select the custom Internet Gateway that we created earlier and **Save changes**.

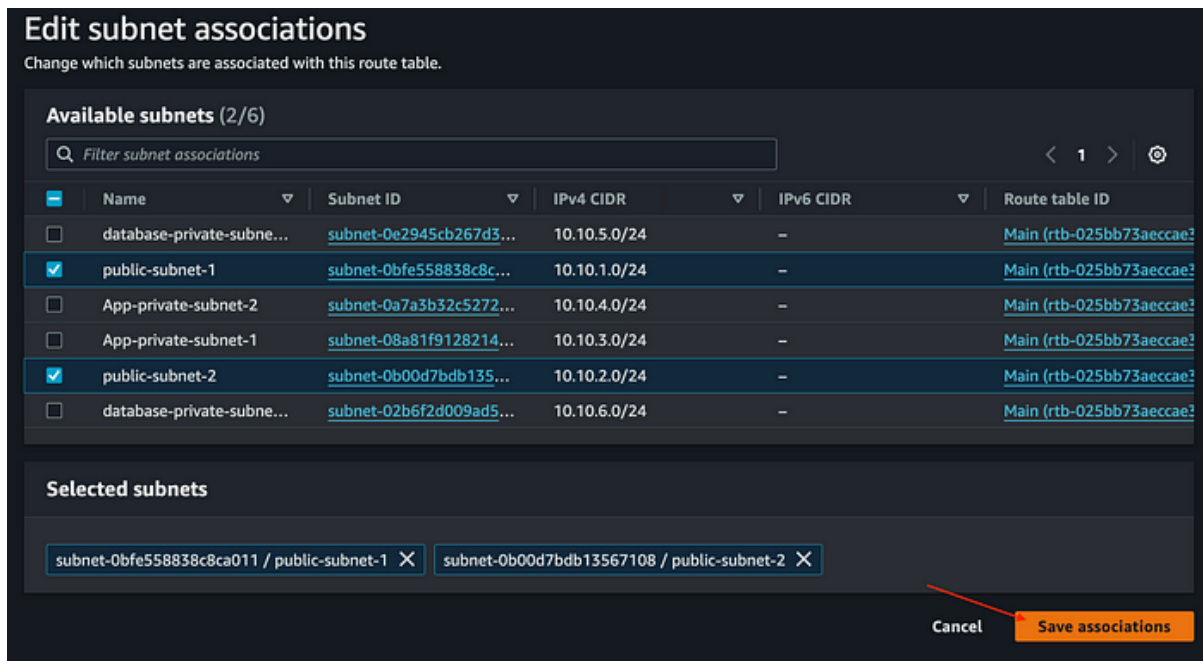
Subnet Association

We now need to explicitly associate our two public subnets with the route table that we just edited. To do this, follow the steps below.

Steps

1. Select **Route tables** in the left pane of the VPC dashboard.
2. Check the box next to the main route table and select the **Subnet associations** tab.

3. Select **Edit subnet associations**.
4. Check the boxes next to the two public subnets we created for our web tier and select **Save associations**.



Edit subnet associations
Change which subnets are associated with this route table.

Available subnets (2/6)

Filter subnet associations

	Name	Subnet ID	IPv4 CIDR	IPv6 CIDR	Route table ID
☐	database-private-subne...	subnet-0e2945cb267d3...	10.10.5.0/24	-	Main (rtb-025bb73aeccae?)
☒	public-subnet-1	subnet-0bfe558838c8c...	10.10.1.0/24	-	Main (rtb-025bb73aeccae?)
☐	App-private-subnet-2	subnet-0a7a3b32c5272...	10.10.4.0/24	-	Main (rtb-025bb73aeccae?)
☐	App-private-subnet-1	subnet-08a81f9128214...	10.10.3.0/24	-	Main (rtb-025bb73aeccae?)
☒	public-subnet-2	subnet-0b00d7bdb135...	10.10.2.0/24	-	Main (rtb-025bb73aeccae?)
☐	database-private-subne...	subnet-02b6f2d009ad5...	10.10.6.0/24	-	Main (rtb-025bb73aeccae?)

Selected subnets

subnet-0bfe558838c8ca011 / public-subnet-1 X subnet-0b00d7bdb13567108 / public-subnet-2 X

Cancel Save associations

As you can see, we now have two subnets explicitly associated with the main route table.

where our network traffic flows and in this case, we want traffic from our private subnets to flow to our NAT gateway. Subnets can only be associated with one route table at a time.

Steps

1. Select **Route tables** in the left pane of the VPC dashboard.
2. Select **Create route table**.
3. Name the route table and select our custom VPC.
4. Select **Create route table**.
5. Once created, select **Edit routes** > **Add route**.
6. Choose **0.0.0.0/0** in the Destination dropdown list. This represents all IPv4 addresses.
7. Select **NAT Gateway** in the Target dropdown list and then select the custom NAT Gateway in AZ1 that we created earlier and **Save changes**.

Edit routes

Route 1

Destination	Target	Status
10.10.0.0/16	local	Active

Propagated: No

Route 2

Destination	Target	Status
0.0.0.0/0	NAT Gateway	-

Propagated: No

Search for NAT Gateway

nat-03a4beb46ad906850 (NAT-GW-AZ1)

nat-0c8e1d9d3268229a6 (NAT-GW-AZ2)

nat-03a4beb46ad906850 (NAT-GW-AZ1)

Remove

Add route

Cancel Preview **Save changes**

We now need to explicitly associate our private subnet in our Application Tier with the route table that we just edited. To do this, follow the steps below.

Steps

1. Select **Route tables** in the left pane of the VPC dashboard.
2. Check the box next to the private route table and select the **Subnet associations** tab.
3. Select **Edit subnet associations**.
4. Check the boxes next to the private subnet in AZ1 for our Application Tier and select **Save associations**.

Edit subnet associations

Change which subnets are associated with this route table.

Available subnets (1/6)

Filter subnet associations

< 1 > ⚙

☐	Name	Subnet ID	IPv4 CIDR	IPv6 CIDR	Route table ID
☐	database-private-subne...	subnet-0e2945cb267d3...	10.10.5.0/24	-	Main (rtb-025bb73aeccae3)
☐	public-subnet-1	subnet-0bfe558838c...	10.10.1.0/24	-	rtb-025bb73aeccae3ff3
☐	App-private-subnet-2	subnet-0a7a3b32c5272...	10.10.4.0/24	-	Main (rtb-025bb73aeccae3)
☒	App-private-subnet-1	subnet-08a81f9128214...	10.10.3.0/24	-	Main (rtb-025bb73aeccae3)
☐	public-subnet-2	subnet-0b00d7bdb135...	10.10.2.0/24	-	rtb-025bb73aeccae3ff3
☐	database-private-subne...	subnet-02b6f2d009ad5...	10.10.6.0/24	-	Main (rtb-025bb73aeccae3)

Selected subnets

subnet-08a81f9128214c43b / App-private-subnet-1 ✕

Cancel

Save associations

As you can see, we now have two subnets explicitly associated with the main route table.

rtb-029b379f0e1680dcb / APP-PRIVATE-ROUTE-TABLE-AZ1

Actions ▾

Details Info

Route table ID

rtb-029b379f0e1680dcb

VPC

vpc-08b90d787f5da9326

3tierVPC

Main

No

Owner ID

381492304791

Explicit subnet associations

subnet-08a81f9128214c43b / App-private-subnet-1

Edge associations

–

Routes

Subnet associations

Edge associations

Route propagation

Tags

Explicit subnet associations (1)

Edit subnet associations

Find subnet association

< 1 > ⚙

Name ▾	Subnet ID ▾	IPv4 CIDR ▾	IPv6 CIDR ▾
App-private-subnet-1	subnet-08a81f9128214...	10.10.3.0/24	–

Subnets without explicit associations (3)

Edit subnet associations

The following subnets have not been explicitly associated with any route tables and are therefore associated with the main route table:

Find subnet association

< 1 > ⚙

Name ▾	Subnet ID ▾	IPv4 CIDR ▾	IPv6 CIDR ▾
database-private-subn...	subnet-0e2945cb267d3...	10.10.5.0/24	–
App-private-subnet-2	subnet-0a7a3b32c5272...	10.10.4.0/24	–
database-private-subn...	subnet-02b6f2d009ad5...	10.10.6.0/24	–

Repeat the same process for the second Application Tier route table and make sure to select the NAT Gateway in the second Availability Zone. Also ensure that it is explicitly associated with the Application Tier private subnet we created in the second Availability Zone in **us-east-1b**.

Edit routes

Route 1

Destination	Target	Status
10.10.0.0/16	local	Active

Propagated: No

Route 2

Destination	Target	Status
0.0.0.0/0	NAT Gateway	-

Propagated: No

nat-0c8e1d9d3268229a6

Use: "nat-0c8e1d9d3268229a6"

nat-0c8e1d9d3268229a6 (NAT-GW-AZ2)

nat-03a4beb46ad906850 (NAT-GW-AZ1)

nat-0c8e1d9d3268229a6 (NAT-GW-AZ2)

Remove

Add route

Cancel Preview Save changes

This same process above will be created for the remaining two route tables for our Database Tier.

Here's what our VPC Resource map looks like.



Web Tier Creation (Tier 1)

Now that we have configured our network and subnets, we will focus on creating the first tier of our 3-tier architecture which is the web tier. We will create an Auto Scaling Group in each AZ with a minimum of 2 instances and a maximum of five. A web server security group along with a custom AMI will be created with the specifications below. Select the **Auto Scaling Groups** in the left pane of the EC2 dashboard and select **Create Auto Scaling group**.

AMI Configuration:

- OS: Amazon Linux 2
- Instance type: t2.micro.
- Create a new Key pair or select an existing one.
- VPC: Select the custom VPC we created earlier.
- Create a security group with SSH, HTTP, and HTTPS inbound rules from anywhere.
- All other options will be left to their default settings.

Static web page configuration:

Expand the **Advanced Details** section, navigate to the **User data** section, and insert the code below. This will install Apache web server software and display a static web page.


```
#!/bin/bash
# Use this for your user data (script from top to bottom)
# install httpd (Linux 2 version)
yum update -y
yum install -y httpd
systemctl start httpd
systemctl enable httpd
echo "<h1>Elite Electronics</h1>" > /var/www/html/index.html
```

After entering the code into the User data section, select **Create launch template**. Let's go back to our Auto Scaling Group setup page and select the new launch template we just created.

Choose launch template [Info](#)

Specify a launch template that contains settings common to all EC2 instances that are launched by this Auto Scaling group.

Name

Auto Scaling group name

Enter a name to identify the group.

WEB-TIER-ASG

Must be unique to this account in the current Region and no more than 255 characters.

Launch template [Info](#)

 For accounts created after May 31, 2023, the EC2 console only supports creating Auto Scaling groups with launch templates. Creating Auto Scaling groups with launch configurations is not recommended but still available via the CLI and API until December 31, 2023.

Launch template

Choose a launch template that contains the instance-level settings, such as the Amazon Machine Image (AMI), instance type, key pair, and security groups.

WEB-TIER-LAUNCH-TEMPLATE ▼



[Create a launch template](#) 

Version

Default (1) ▼



[Create a launch template version](#) 

Description

WEB-TIER-LAUNCH-TEMPLATE

Launch template

[WEB-TIER-LAUNCH-TEMPLATE](#) 

lt-0fff7f42bc6bc70ef

Select **Next** and choose the custom VPC we created and select the two public subnets under **Availability Zones and subnets** and select **Next**.

Network [Info](#)

For most applications, you can use multiple Availability Zones and let EC2 Auto Scaling balance your instances across the zones. The default VPC and default subnets are suitable for getting started quickly.

VPC
Choose the VPC that defines the virtual network for your Auto Scaling group.

vpc-08b90d787f5da9326 (3tierVPC) 10.10.0.0/16

[Create a VPC](#)

Availability Zones and subnets
Define which Availability Zones and subnets your Auto Scaling group can use in the chosen VPC.

Select Availability Zones and subnets

us-east-1a | subnet-0bfe558838c8ca011 (public-subnet-1) 10.10.1.0/24

us-east-1b | subnet-0b00d7bdb13567108 (public-subnet-2) 10.10.2.0/24

[Create a subnet](#)

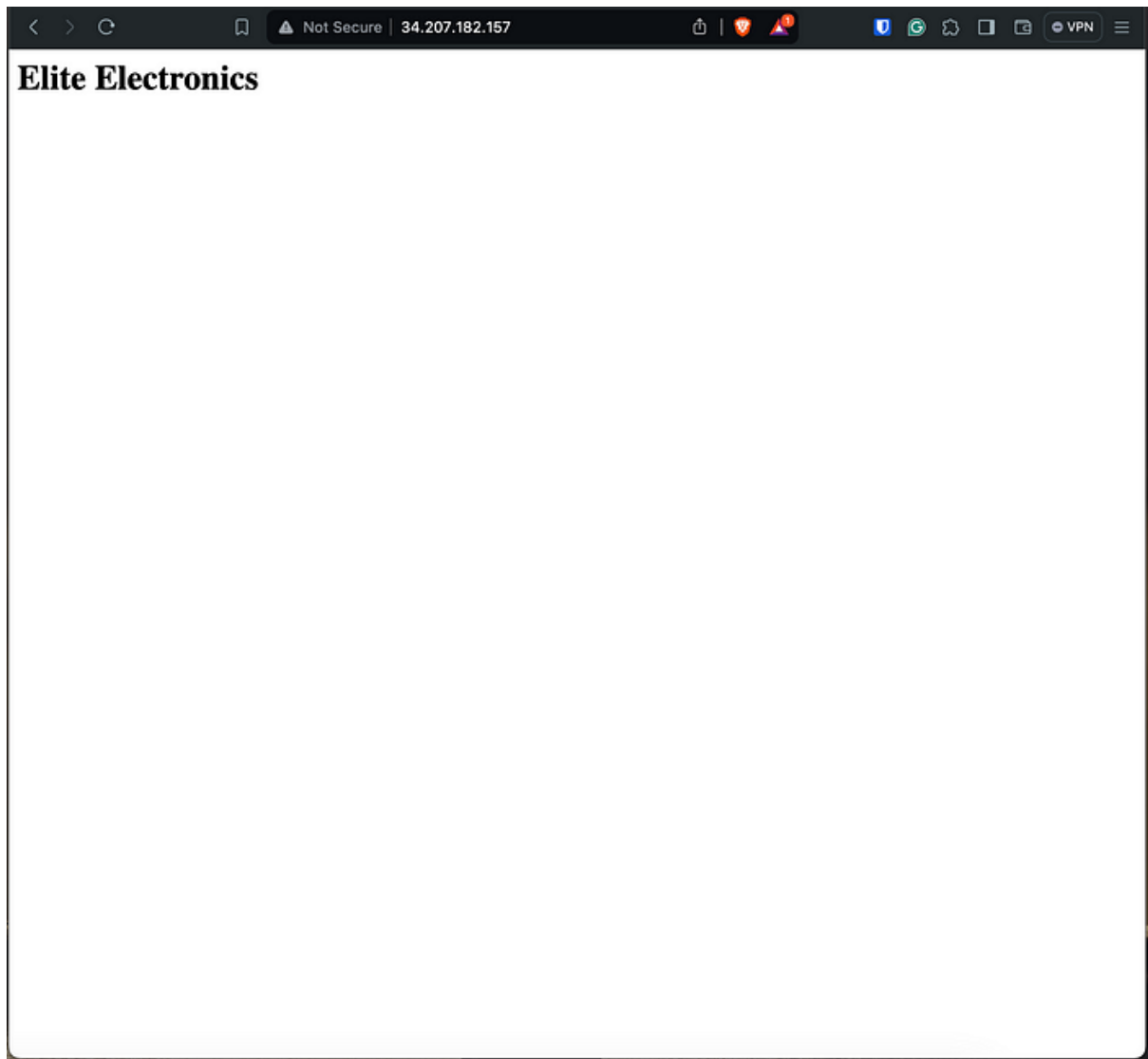
Cancel Skip to review Previous **Next**

We will not configure a load balancer at this time. Select **Next**. For our group size and scaling options, we will configure with the values below.

- Desired capacity: 2
- Min desired capacity: 2
- Max desired capacity: 5
- Automatic scaling: Choose **Target tracking scaling policy**

- We will leave the Average CPU utilization target value at 50
- Select **Next** until you arrive at the **Review** page. Select **Create Auto Scaling group** and wait for your instances to launch.

After our two instances are up and running, we want to verify that we can reach the static web page from the instance's public IP address. Select one instance at a time, copy and paste the public IPv4 address into your browser window, and see if we can access the static web page.



Application Tier (Tier 2)

We will now focus on building out the second tier of our 3-tier architecture which is the Application Tier. The Application Tier will also include an auto scaling group along with an EC2 Application Server Security Group allowing inbound permission from the Web Server Security Group. A custom AMI will be created with the specifications below. Select the **Auto Scaling Groups** in the left

pane of the EC2 dashboard and select **Create Auto Scaling group**.

Attention: This is only a demo application tier because we don't have any code to run on the EC2 instances.

AMI Configuration:

- OS: Amazon Linux 2 Kernel 5.10 AMI 2.0.20240521.0 x86_64 HVM gp2.
- Instance type: t2.micro.
- Create a new Key pair or select an existing one.
- VPC: Select the custom VPC we created earlier.
- Create a security group with SSH, and ICMP (to ping application instance from web tier) inbound rules from the web server security group.
- All other options will be left to their default settings.

Firewall (security groups) [Info](#)

A security group is a set of firewall rules that control the traffic for your instance. Add rules to allow specific traffic to reach your instance.

☐ Select existing security group

☒ Create security group

Security group name - *required*

APP-TIER-SG

This security group will be added to all network interfaces. The name can't be edited after the security group is created. Max length is 255 characters. Valid characters: a-z, A-Z, 0-9, spaces, and . _ - / () # , @ [] + = & ; ! \$ *

Description - *required* [Info](#)

Allow SSH Access and ICMP to ping from Web Tier

VPC [Info](#)

vpc-08b90d787f5da9326 (3tierVPC)
10.10.0.0/16

Inbound Security Group Rules

▼ Security group rule 1 (TCP, 22, sg-02f7cd87e9fe919cc)

Remove

Type [Info](#)

ssh

Protocol [Info](#)

TCP

Port range [Info](#)

22

Source type [Info](#)

Custom

Source [Info](#)

Use: sg

Security groups

default
sg-0e12df826f5d923d1

launch-wizard-1
sg-05ccb78150dee9590

default
sg-08cb9176ad0ed511b

Web-server-sg
sg-02f7cd87e9fe919cc

Description - *optional* [Info](#)

e.g. SSH for admin desktop

Remove

▼ Security group rule 2 (ICMP, All)

Type [Info](#)

All ICMP - IPv4

Source type [Info](#)

Custom

Port range [Info](#)

All

Description - *optional* [Info](#)

SSH for admin desktop

Add security group rule

▼ Advanced network configuration

No network interfaces are currently included in this template. Add a network interface to include it in the launch template.

Let's go back to our Auto Scaling Group setup page and select the new launch template we just created

Launch template Info

For accounts created after May 31, 2023, the EC2 console only supports creating Auto Scaling groups with launch templates. Creating Auto Scaling groups with launch configurations is not recommended but still available via the CLI and API until December 31, 2023.

Launch template

Choose a launch template that contains the instance-level settings, such as the Amazon Machine Image (AMI), instance type, key pair, and security groups.

APP-TIER-LAUNCH-TEMPLATE

Create a launch template

Version

Default (1)

Create a launch template version

Description

APP-TIER-LAUNCH-TEMPLATE

Instance type

t2.micro

Security groups

-

Key pair name

VPC_KeyPair

Launch template

APP-TIER-LAUNCH-TEMPLATE

lt-0b9d76dcf813cff63

AMI ID

ami-0d94353f7bad10668

Request Spot Instances

No

Security group IDs

sg-0461ec26c60d550fe

Additional details

Storage (volumes)

-

Date created

Sun May 26 2024 16:23:34 GMT-0500 (Central Daylight Time)

Cancel

Next

Select **Next** and choose the custom VPC we created and select the two private subnets for the application tier under **Availability Zones and subnets** and select **Next**.

Instance type requirements

Info

Override launch template

You can keep the same instance attributes or instance type from your launch template, or you can choose to override the launch template by specifying different instance attributes or manually adding instance types.

Launch template	Version
APP-TIER-LAUNCH-TEMPLATE lt-0b9d76dcf813cff63	Default
Description	Instance type
APP-TIER-LAUNCH-TEMPLATE	t2.micro

Network

Info

For most applications, you can use multiple Availability Zones and let EC2 Auto Scaling balance your instances across the zones. The default VPC and default subnets are suitable for getting started quickly.

VPC

Choose the VPC that defines the virtual network for your Auto Scaling group.

vpc-08b90d787f5da9326 (3tierVPC)

10.10.0.0/16

↺

Create a VPC

Availability Zones and subnets

Define which Availability Zones and subnets your Auto Scaling group can use in the chosen VPC.

Select Availability Zones and subnets

↺

us-east-1a | subnet-08a81f9128214c43b

(App-private-subnet-1)

10.10.3.0/24

×

us-east-1b | subnet-0a7a3b32c52721062

(App-private-subnet-2)

10.10.4.0/24

×

Create a subnet

Cancel

Skip to review

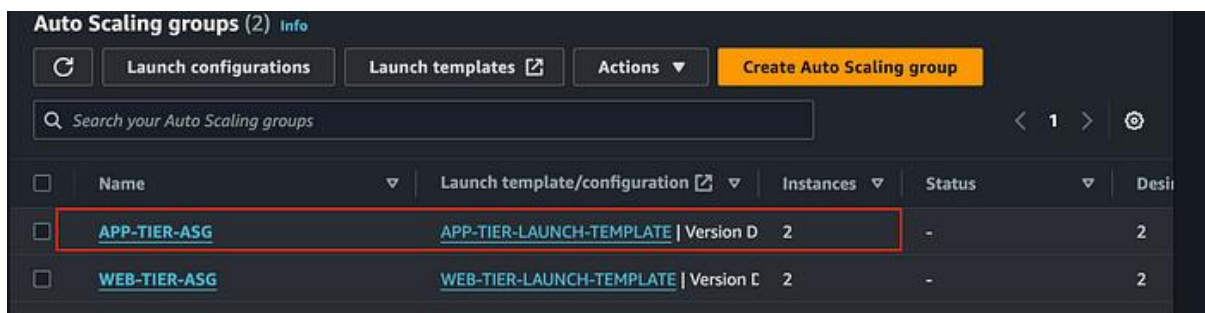
Previous

Next

We will not configure a load balancer at this time. Select **Next**. For our group size and scaling options, we will configure with the values below.

- Desired capacity: 2
- Min desired capacity: 2
- Max desired capacity: 5
- Automatic scaling: Choose **Target tracking scaling policy**
- We will leave the Average CPU utilization target value at 50
- Select **Next** until you arrive at the **Review** page. Select **Create Auto Scaling group** and wait for your instances to launch.

Our Application Tier Auto Scaling Group has been successfully created.



	Name	Launch template/configuration	Instances	Status	Desired capacity
☐	APP-TIER-ASG	APP-TIER-LAUNCH-TEMPLATE Version D	2	-	2
☐	WEB-TIER-ASG	WEB-TIER-LAUNCH-TEMPLATE Version C	2	-	2

Note that our instances in our Application Tier Auto Scaling Group don't have a public IPv4 address assigned. Even though the instances aren't publicly accessible, we still want the web tier to be able to communicate with the application tier so we will attempt to ping an instance in the application tier auto scaling group.

Public IPv4 ...	Elastic IP	IPv6 IPs	Monitoring	Security group name
–	–	–	disabled	APP-TIER-SG
–	–	–	disabled	APP-TIER-SG
34.207.182.157	–	–	disabled	Web-server-sg
54.144.2.228	–	–	disabled	Web-server-sg

So we've successfully verified that the web tier has connectivity to our application tier instances by pinging one of the application servers. What if one of our engineers needs to securely connect to one of our application servers remotely within our private subnet? This is where a bastion host comes into play. A bastion host provides a centralized network access control point to our EC2 instances located in our application tier in our private subnets.

We normally connect remotely to an EC2 instance utilizing a key pair. Connecting to a bastion host in a private subnet requires using a private key that should never be stored on the bastion instance. To solve this problem we will utilize SSH agent forwarding to allow cloud engineers secure, remote access to instances located in our private subnets without storing the private key on the bastion host.

Bastion Host Creation

1. Navigate to the EC2 dashboard and select **Launch instances**.
2. Name your instance, select Amazon Linux 2 for the OS, t2.micro for the instance type. You can create a new key pair or use an existing one.
3. Select **Edit** next to the Network settings to choose your custom VPC we created earlier and select a public subnet. Name the security group and give it a description. Enable **Auto-assign public IP**.
4. Create a new security group and allow SSH traffic from your IP address.

5. Select **Launch** instance.

▼ Network settings Info

VPC - required Info

vpc-08b90d787f5da9326 (3tierVPC)
10.10.0.0/16

↻

Subnet Info

subnet-0bfe558838c8ca011 public-subnet-1
VPC: vpc-08b90d787f5da9326 Owner: 381492304791 Availability Zone: us-east-1a
IP addresses available: 249 CIDR: 10.10.1.0/24

↻ Create new subnet

Auto-assign public IP Info

Enable

Additional charges apply when outside of free tier allowance

Firewall (security groups) Info

A security group is a set of firewall rules that control the traffic for your instance. Add rules to allow specific traffic to reach your instance.

☒ Create security group ☐ Select existing security group

Security group name - required

Bastion-Host-SG

This security group will be added to all network interfaces. The name can't be edited after the security group is created. Max length is 255 characters. Valid characters: a-z, A-Z, 0-9, spaces, and ._-:/()#,@[]+=&;{}!\$*

Description - required Info

Bastion Host SG

Inbound Security Group Rules

▼ Security group rule 1 (TCP, 22, 76.31.123.77/32) Remove

Type Info	Protocol Info	Port range Info
ssh	TCP	22
Source type Info	Name Info	Description - optional Info
My IP	🔍 Add CIDR, prefix list or security group 76.31.123.77/32 ✕	e.g. SSH for admin desktop

Add security group rule

► Advanced network configuration

Note: Now that we have set up our bastion host to allow SSH into our instances in the application tier, we need to change the inbound rules in the Application tier security group to allow SSH access from the Bastion security group we just created and remove the SSH inbound rule from the Web Tier security group.

Edit inbound rules Info

Inbound rules control the incoming traffic that's allowed to reach the instance.

Inbound rules Info

Inbound rule 1 Delete

Security group rule ID
sgr-014e4f61432b574aa

Type Info
SSH

Protocol Info
TCP

Port range Info
22

Source type Info
Custom

Source Info
 ×

Security Groups
Bastion-Host-SG | sg-0cd7624540194e967

Description - optional Info

Bastion-Host-SG | sg-0cd7624540194e967

Inbound rule 2 Delete

Security group rule ID
sgr-0efd12d5ec84a54d8

Type Info
All ICMP - IPv4

Protocol Info
ICMP

Port range Info
All

Source type Info
Custom

Source Info
 ×

Description - optional Info

Add rule

Cancel

Preview changes

Save rules

Now that we've changed our inbound rules, we can connect to our instances in our private subnet hosting our application tier with the following command:

Database Tier Creation (3rd Tier)

For the database tier, we will create one MySQL database instance in one private subnet. I will document how to create a Multi-AZ database configuration later on.

Subnet Groups

Before we create our database, we need to specify what private subnets our database will utilize by creating a subnet group. This will allow us to select this custom subnet group during our database creation. To create a subnet group, follow the steps below.

Steps

1. Navigate to the RDS dashboard and select **Subnet groups** in the left pane.
2. Select **Create DB subnet group**.
3. Name your subnet group and give it a description.
4. Select the custom VPC we created and under Availability Zones, choose the two AZs that we deployed our resources to. For me, it was **us-east-1a** and **us-east-1b**.

5. Select the two database subnets that we created earlier under the Subnets option. For Multi-AZ DB clusters you must select 3 subnets in 3 different AZs.
6. Select **Create**.

Create DB subnet group

To create a new subnet group, give it a name and a description, and choose an existing VPC. You will then be able to add subnets related to that VPC.

Subnet group details

Name

You won't be able to modify the name after your subnet group has been created.

My-DB-Subnet-Group

Must contain from 1 to 255 characters. Alphanumeric characters, spaces, hyphens, underscores, and periods are allowed.

Description

My-DB-Subnet-Group

VPC

Choose a VPC identifier that corresponds to the subnets you want to use for your DB subnet group. You won't be able to choose a different VPC identifier after your subnet group has been created.

3tierVPC (vpc-08b90d787f5da9326)

Add subnets

Availability Zones

Choose the Availability Zones that include the subnets you want to add.

Choose an availability zone

us-east-1a

us-east-1b

Subnets

Choose the subnets that you want to add. The list includes the subnets in the selected Availability Zones.

Select subnets

subnet-0e2945cb267d36d40 (10.10.5.0/24)

subnet-02b6f2d009ad560e6 (10.10.6.0/24)

i For Multi-AZ DB clusters, you must select 3 subnets in 3 different Availability Zones.

Subnets selected (2)

Availability zone	Subnet ID	CIDR block
us-east-1a	subnet-0e2945cb267d36d40	10.10.5.0/24
us-east-1b	subnet-02b6f2d009ad560e6	10.10.6.0/24

Cancel

Create

Database Creation

Steps

1. Navigate to the RDS dashboard and select **Create database**.
2. Select **Standard create** and **MySQL** for the Engine options.
3. Select **Free tier** under Templates. If you wanted to deploy your database in a Multi-AZ configuration, you would need to select the **Production** or **Dev/Test** option.
4. Select Single DB instance under Availability and durability. If you were deploying in a Multi-AZ configuration you would choose either the **Multi-AZ DB Cluster** or **Multi-AZ DB instance** options.
5. Name your database, you can use the default Master username or you can change it your own, select the **Self managed** option under **Credentials Settings** and make up your own password.
6. For the Instance configuration, we'll use the **Burstable classes** option and select the **db.t3.micro** option.
7. Leave the Storage settings as default.
8. Under **Connectivity**, leave the **Don't connect to an EC2 compute resource** option selected, choose the custom VPC we created earlier, choose the DB subnet group we just created under **DB subnet group** and leave the **Public access** option to the **No** setting.

9. Under **VPC security group**, select **Create new**, name the security group. If you expand the **Additional configuration** section, the Database port is set to allow connections on port **3306**.

The screenshot shows the 'Create new database' page in the AWS RDS console. The 'VPC security group (firewall)' section is active, with the 'Create new' option selected. The 'New VPC security group name' field contains 'DB-SG'. The 'Availability Zone' is set to 'No preference'. The 'RDS Proxy' section is collapsed. The 'Certificate authority - optional' section is expanded, showing 'rds-ca-rsa2048-g1 (default)' as the selected certificate authority. The 'Additional configuration' section is expanded, showing the 'Database port' set to '3306'.

VPC security group (firewall) [Info](#)
Choose one or more VPC security groups to allow access to your database. Make sure that the security group rules allow the appropriate incoming traffic.

☐ Choose existing
Choose existing VPC security groups

☒ Create new
Create new VPC security group

New VPC security group name
DB-SG

Availability Zone [Info](#)
No preference

RDS Proxy
RDS Proxy is a fully managed, highly available database proxy that improves application scalability, resiliency, and security.

☐ Create an RDS Proxy [Info](#)
RDS automatically creates an IAM role and a Secrets Manager secret for the proxy. RDS Proxy has additional costs. For more information, see [Amazon RDS Proxy pricing](#).

Certificate authority - optional [Info](#)
Using a server certificate provides an extra layer of security by validating that the connection is being made to an Amazon database. It does so by checking the server certificate that is automatically installed on all databases that you provision.

rds-ca-rsa2048-g1 (default)
Expiry: May 25, 2061

If you don't select a certificate authority, RDS chooses one for you.

▼ **Additional configuration**

Database port [Info](#)
TCP/IP port that the database will use for application connections.

3306

10. Leave all other settings as default and select **Create database**.

Once our database is created, we want to allow inbound traffic from the Application Server Security Group.

Steps

1. Select the new database and choose the **Connectivity & security** tab and select the database security group we created earlier.

The screenshot displays the AWS RDS console interface for a database instance named 'database-1'. At the top, there are buttons for 'Refresh', 'Modify', and 'Actions'. Below this is a 'Summary' section with a table of instance details:

DB identifier	Status	Role	Engine	Recommendations
database-1	⌚ Backing-up	Instance	MySQL Community	
CPU 44.02%	Class db.t3.micro	Current activity 0	Region & AZ us-east-1b	

Below the summary is a navigation bar with tabs: 'Connectivity & security' (selected), 'Monitoring', 'Logs & events', 'Configuration', 'Zero-ETL integrations', and 'Maintenance'. The 'Connectivity & security' tab is active, showing three columns: 'Endpoint & port', 'Networking', and 'Security'.

- Endpoint & port:** Endpoint is 'database-1.cre62ugcavwh.us-east-1.rds.amazonaws.com', and Port is '3306'.
- Networking:** Availability Zone is 'us-east-1b', VPC is '3tierVPC (vpc-08b90d787f5da9326)', Subnet group is 'my-db-subnet-group', and Subnets are 'subnet-0e2945cb267d36d40' and 'subnet-02b6f2d009ad560e6'. Network type is 'IPv4'.
- Security:** VPC security groups list shows 'DB-SG (sg-04d3e4214df1540f6)' with a red arrow pointing to it and a green 'Active' status. 'Publicly accessible' is set to 'No'. Certificate authority is 'rds-ca-rsa2048-g1'. Certificate authority date is 'May 25, 2061, 18:34 (UTC-05:00)'. DB instance certificate expiration date is 'May 26, 2025, 19:54 (UTC-05:00)'.

2. Select the **Inbound rules** tab and then select **Edit inbound rules**.

Edit inbound rules Info

Inbound rules control the incoming traffic that's allowed to reach the instance.

Inbound rules Info

Security group rule ID	Type Info	Protocol Info	Port range Info	Source Info	Description - optional Info	
-	MySQL/Aurora	TCP	3306	Cus...	sg-0461ec26c60d550fe	sg-0461ec26c60d550fe X

Add rule

Cancel

Preview changes

Save rules

Inbound security group rules successfully modified on security group sg-04d3e4214df1540f6 (DB-SG)

Details

Security Groups (1/7) Info

Find resources by attribute or tag

Name	Security group ID	Security group name	VPC ID	Description	Owner	Inbound rules count	Outbound rules count
-	sg-04d3e4214df1540f6	DB-SG	vpc-08b00c78775da9326	Created by RDS management console	381492304791	1 Permission entry	1 Permission entry
-	sg-0a12a0f92f4d92161	default	vpc-08b00c78775da9326	default VPC security group	381492304791	1 Permission entry	1 Permission entry
-	sg-05c0b781509ec5990	launch-wizard-1	vpc-08b00c78775da9326	launch-wizard-1 created 2024-04-08T...	381492304791	2 Permission entries	1 Permission entry
-	sg-0c07624540194c967	bastion-host-SG	vpc-08b00c78775da9326	Bastion Host SG	381492304791	1 Permission entry	1 Permission entry
-	sg-08b00c78775da9326	default	vpc-08b00c78775da9326	default VPC security group	381492304791	1 Permission entry	1 Permission entry
-	sg-0461ec26c60d550fe	APP-TIER-SG	vpc-08b00c78775da9326	Allow SSH Access and ICMP to ping fro...	381492304791	2 Permission entries	1 Permission entry

sg-04d3e4214df1540f6 - DB-SG

Details Inbound rules Outbound rules Tags

Inbound rules (1)

Search

Name	Security group rule...	IP version	Type	Protocol	Port range	Source	Description
-	sg-0461ec26c60d550fe	-	MySQL/Aurora	TCP	3306	sg-0461ec26c60d550fe / APP-TIER-SG	-

Now that we've got our foundational 3-tier architecture created, let's create a load balancer for the web tier to evenly distribute the load to our EC2 instances.

Steps

1. Navigate to the EC2 dashboard and select **Load Balancers** in the left pane. Select **Create load balancer**.
2. Select **Create** under Application Load Balancer.
3. Name load balancer, scheme is **Internet-facing**, IP address type is **IPv4**.

4. Select the custom VPC we created earlier and the two public subnets we created in the two different AZs such as us-east-1a and us-east-1b.
5. Delete the default security group and Select **Create new security group**. Name the security group, assign it to your custom VPC and select **Add rule** under Inbound rules.
6. Set HTTP access on port 80 from Anywhere 0.0.0.0/0 and select **Create security group**.
7. Let's go back to our Load Balancer page and refresh the Security groups. Remove the default security group and select the new security group we just created for our load balancer.

Security groups [Info](#)
A security group is a set of firewall rules that control the traffic to your load balancer. Select an existing security group, or you can [create a new security group](#).

Security groups
Select up to 5 security groups

Security group	VPC
☒ Web-alb-sg sg-00d2b2dfab94f9d7d	vpc-08b90d787f5da9326
☐ DB-SG sg-04d3e4214df1540f6	vpc-08b90d787f5da9326
☐ Bastion-Host-SG sg-0cd7624540194e967	vpc-08b90d787f5da9326
☐ default sg-08cb9176ad0ed511b	vpc-08b90d787f5da9326
☐ APP-TIER-SG sg-0461ec26c60d550fe	vpc-08b90d787f5da9326
☐ Web-server-sg sg-02f7cd87e9fe919cc	vpc-08b90d787f5da9326

Protocol: HTTP **Port**: 80 **Default action**: [Info](#)
Forward to: [Select a target group](#) [Create target group](#)

Listener tags - optional
Consider adding tags to your listener. Tags enable you to categorize your AWS resources so you can more easily manage them.
[Add listener tag](#)
You can add up to 50 more tags.

[Add listener](#)

Create Target Group

We need to have instances to direct our traffic to coming from the load balancer. Now we will create a target group.

Steps

1. Under the **Listeners and routing** section, select **Create target group**.
2. Leave Target type as **Instances** and give your target group a name.
3. Leave all other options as default and select **Next**.
4. Select the two instances from our web tier and select **Include as pending below** to register the targets.

Register targets
This is an optional step to create a target group. However, to ensure that your load balancer routes traffic to this target group you must register your targets.

Available instances (2/5)

Filter instances

Instance ID	Name	State	Security groups	Zone	Private IPv4 address	Subnet ID	Launch time
i-02011afabb99fbd1	Bastion Host	running	Bastion-Host-SG	us-east-1a	10.10.1.48	subnet-0bf4558838c8ca011	May 26, 2024, 17:44 (UTC-05:00)
i-0630c8da17b4ae84c		running	APP-TIER-SG	us-east-1b	10.10.4.44	subnet-0a7a3b32c52721062	May 26, 2024, 16:31 (UTC-05:00)
i-0838c9ba95a88a3f		running	APP-TIER-SG	us-east-1a	10.10.3.158	subnet-0ba81f9128214c43b	May 26, 2024, 16:31 (UTC-05:00)
i-00d099611886722ab		running	Web-server-sg	us-east-1b	10.10.2.186	subnet-0b00d70db13567108	May 26, 2024, 15:53 (UTC-05:00)
i-02447b00a1851f83		running	Web-server-sg	us-east-1a	10.10.1.195	subnet-0bf4558838c8ca011	May 26, 2024, 15:53 (UTC-05:00)

2 selected

Ports for the selected instances
Ports for routing traffic to the selected instances.

80

1-65535 (separate multiple ports with comma)

Include as pending below

Targets for Web Tier ALB successfully registered.

Targets (2) Remove all pending

Show only pending

Instance ID	Name	Port	State	Security groups	Zone	Priv
i-00d0b9611886722ab		80	Running	Web-server-sg	us-east-1b	10.
i-02447bb00a1851f83		80	Running	Web-server-sg	us-east-1a	10.

2 pending Cancel Previous Create target group

5. Select **Create target group**.

6. Once our target group is created, navigate back to the Load Balancer page, refresh the target group section, and choose the target group we just created.

Listeners and routing Info

A listener is a process that checks for connection requests using the port and protocol you configure. The rules that you define for a listener determine how the load balancer routes requests to its registered targets.

▼ Listener HTTP:80 Remove

Protocol: HTTP Port: 80 (1-65535)

Default action: Forward to **WEB-ALB-TG** HTTP Refresh

Target type: Instance, IPv4

[Create target group](#)

Listener tags - optional

Consider adding tags to your listener. Tags enable you to categorize your AWS resources so you can more easily manage them.

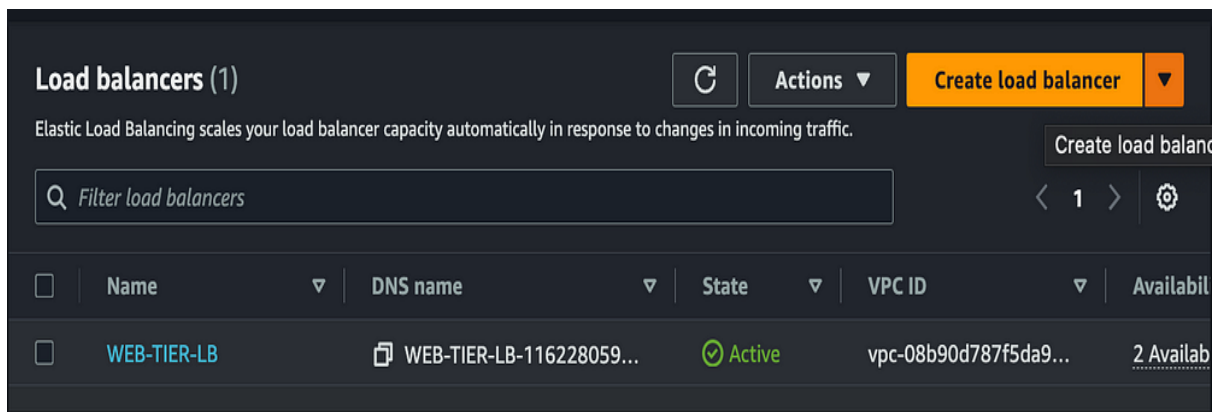
Add listener tag

You can add up to 50 more tags.

Add listener

7. Scroll down to the bottom of the page and select **Create load balancer**.

Our new load balancer for our Web Tier is created.



Since Health Checks were enabled on our target group, we can see below that both of our instances are healthy.

WEB-ALB-TG

Actions

Details

arn:aws:elasticloadbalancing:us-east-1:381492304791:targetgroup/WEB-ALB-TG/74956fa846278014

Target type

Instance

IP address type

IPv4

Protocol : Port

HTTP: 80

Load balancer

None associated

Protocol version

HTTP1

VPC

vpc-08b90d787f5da9326

2

Total targets

2

Healthy

0 Anomalous

0

Unhealthy

0

Unused

0

Initial

0

Draining

Distribution of targets by Availability Zone (AZ)

Select values in this table to see corresponding filters applied to the Registered targets table below.

Targets

Monitoring

Health checks

Attributes

Tags

Registered targets (2)

Info

Anomaly mitigation: Not applicable

Deregister

Register targets

Target groups route requests to individual registered targets using the protocol and port number specified. Health checks are performed on all registered targets according to the target group's health check settings. Anomaly detection is automatically applied to targets that are not healthy. Refresh Registered targets

Filter targets

1

Instance ID

Name

Port

Zone

Health status

Health s

i-02447bb00a1851f83

80

us-east-1a

Healthy

-

i-00d0b9611886722ab

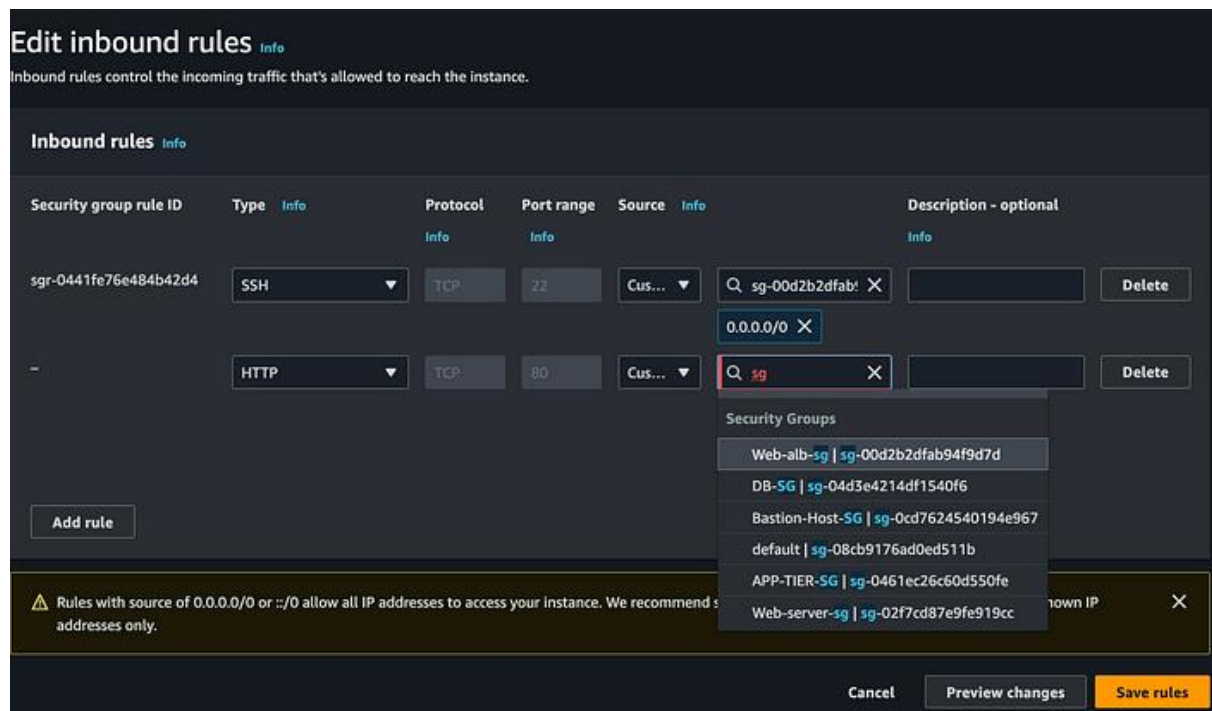
80

us-east-1b

Healthy

-

The ALB should allow HTTP traffic from 0.0.0.0/0 and the Web Server Security group should only allow traffic only from the ALB. We need to select the Web Tier security group and edit the inbound rules. If you get an error, delete the current rule and add another rule to enable HTTP traffic on port 80.



Summary

We've successfully created a 3-tier architecture in AWS. Below is a quick recap of what we've accomplished:

- Created 1 public subnet and two private subnets
- Created and IGW and ALB for our web tier
- Created an autoscaling group for our web and application tiers
- Created security groups
- Created a Bastion Host to allow private access to our application tier
- Created NAT Gateways to allow our EC2 instances in our application tier internet access.

- Created a MySQL RDS database that allows for easy scaling